Routed Experts Are Not Redundant: The Shared-Expert Gradient Race in Small Mixture-of-Experts Models

TBD

Abstract

Small Mixture-of-Experts (MoE) language models with a DeepSeek-style shared expert are a popular testbed for interpretability, because they are cheap to train and every token can be labelled. A recurring observation in this regime is that the routed experts appear *redundant*: ablating the entire routed branch barely changes perplexity, while the single always-on shared expert dominates the feed-forward computation. We show that this "routed redundancy" is largely an **artifact of undertraining**, and we identify its cause. Using a 134M-parameter MoE trained from scratch on a three-domain, token-labelled corpus, we find that (i) the shared expert carries ∼80% of the feed-forward ablation impact at 98M tokens but the routed and shared branches become **balanced by ∼300M tokens** and stay balanced through 1B (4 seeds, tight error bars); (ii) this early dominance is a **gradient-exposure race**—the shared expert receives gradient from 100% of tokens versus $\sim k/E$ per routed expert, so it wins the early optimization race; and (iii) the race is **causally controllable**: slowing the shared expert's learning rate shifts the routed-activation crossover dose-dependently (the shared branch's share of ablation impact at 98M tokens drops from 0.80 to 0.14) **at no perplexity cost**. A single hyperparameter thus reallocates computation to the routed experts without changing model quality. The control **reproduces at 2× depth**, though within our token budget the crossover to routed balance completes only at the smallest configuration. We further report the axes of specialization (surface form > domain > frequency, position irrelevant) and a non-monotonic dependence on the number of experts. Two of our own early conclusions were overturned by seed replication and are reported as such.

1 Introduction

Mixture-of-Experts layers replace a dense feed-forward block with a router that dispatches each token to a small number of experts. The DeepSeekMoE design (2) adds an always-on *shared expert* alongside the *routed* experts, on the intuition that the shared expert absorbs common computation and frees the routed experts to specialize. Because small MoEs are cheap and controllable, they are widely used to study expert specialization.

In this small-scale regime, a striking and repeatedly observed phenomenon is that the routed experts look nearly useless: removing all of them costs little perplexity, while removing the lone shared expert costs a lot. It is tempting to conclude that small MoEs are over-parametrized on the routed side, or that the shared expert suppresses the routed experts.

We show that this conclusion is **premature**. The apparent redundancy is a transient property of early training that disappears with enough tokens, and its cause is a simple, controllable optimization dynamic. Our contributions:

1. **Observation.** Branch importance is a function of tokens seen: routed "redundancy" and shared dominance vanish by ∼300M tokens (Section 4).

2. **Mechanism.** The dominance is a *gradient-exposure race*—the shared expert is updated on every token, each routed expert on a k/E fraction—proven causally by slowing the shared expert and watching the crossover shift, at no quality cost (Section 5).

3. **Control and scope.** One hyperparameter (shared-expert learning rate) sets when/whether the routed experts engage; the dependence on the number of experts is non-monotonic, not the clean law the mechanism naively predicts (Section 6).

Along the way we characterize *what* the experts specialize by, and we explicitly retract two early findings that did not survive replication (Section 7). We view the honest, seed-replicated, causal methodology as part of the contribution.

2 Related work

Sparse MoE layers. Conditional computation via a sparsely-gated MoE layer was introduced by Shazeer et al. (14) and scaled by GShard (9) and Switch Transformer (5), which also established the auxiliary load-balancing loss. ST-MoE (17) added the router z-loss for stability. Our model uses both, in a top-2 configuration (8).

Shared experts. DeepSeekMoE (2) introduced the always-on shared expert alongside fine-grained routed experts, arguing it absorbs common knowledge and thereby *frees* the routed experts to specialize; the design carries through DeepSeek-V2/V3 (3, 4), which also motivate auxiliary-loss-free routing by the balance–specialization tension. We reproduce the "shared enables specialization" effect (Section 7) but show it is only visible once balancing is off, and—more importantly—that at small scale the shared expert *dominates* the computation for a mechanistic reason unrelated to specialization.

Expert specialization studies. Analyses of Mixtral (8) report that experts specialize more by token/syntactic identity than by topic; OLMoE (10) studies specialization and routing across training in open MoEs. We add a controlled, domain-labelled probe with causal branch ablation and, crucially, identify a training-dynamics cause for the small-scale "routed redundancy" that these observational studies do not address.

Architecture components. Our backbone uses MLA (3), RoPE (15), RMSNorm (16), and optional YaRN context extension (13); the distillation experiment follows Hinton et al. (6) with a Mistral-7B teacher (7). Pre-training data are FineWeb-Edu (12), Cosmopedia-v2 (1), and OpenWebMath (11).

To our knowledge, the central claim—that small-MoE routed redundancy is an *undertraining artifact* caused by a *gradient-exposure race* between the shared and routed experts, and is *causally controllable* by the shared-expert learning rate at no quality cost—is new.

3 Setup

Model. A decoder-only MoE re-implemented from scratch in PyTorch: Multi-head Latent Attention (MLA), RoPE, RMSNorm, and per layer a MoE block with 8 routed experts + 1 shared expert and top-2 routing (134M parameters, 8 layers, hidden 512, context 512). During analysis we run the MoE in inference mode (no capacity dropping), which we verified is numerically identical to the training MoE when no tokens are dropped.

Table 1: Branch importance vs. tokens (baseline, 4 seeds, batch 64).

tokens	PPL	remove shared	remove all routed	routed NMI	shared share
98M	4.18	$+3.44 \pm 0.17$	$+0.82 \pm 0.03$	0.0121	0.81
197M	3.63	$+2.37 \pm 0.17$	$+1.58 \pm 0.06$	0.0148	0.60
295M	3.41	$+1.99 \pm 0.14$	$+1.98 \pm 0.06$	0.0161	0.50
491M	3.20	$+1.99 \pm 0.16$	$+2.25 \pm 0.01$	0.0168	0.47
983M	3.00	$+2.45 \pm 0.18$	$+2.48 \pm 0.11$	0.0174	0.50

Data. Pre-training streams three token-labelled corpora: FineWeb-Edu (`web`, 60%), Cosmopedia-v2 (`edu`, 25%), OpenWebMath (`math`, 15%), tokenized with the Mistral vocabulary (32k). Sequences are packed to 512 tokens. The domain labels are the key lever: they let us attribute routing to a known variable. Held-out per-domain evaluation packs each domain separately (one label per sequence).

Metrics. *Branch importance*: ΔPPL from ablating a whole branch (all routed experts, or the shared expert). This whole-branch ablation is the correct causal probe—single-expert ablation is misleading because top-2 routing reroutes to the survivors. The *shared share* is $\Delta_{\text{shared}}/(\Delta_{\text{shared}} + \Delta_{\text{routed}})$. *Specialization*: gate-weighted normalized mutual information $\text{NMI}(\text{Expert}; \text{axis}) = I/H$, averaged over layers, against a label-permutation null (≈ 0 everywhere). *Balance*: std of top-1 expert load. Between-condition tests are exact permutation tests on seed-level values (minimum two-sided $p = 2/\binom{2n}{n}$ for n seeds per group).

Compute. All runs are on one NVIDIA RTX A6000 using a plain single-GPU trainer; the largest models see $\sim$1B tokens ($\sim$3.4h each).

4 Routed "redundancy" is an undertraining artifact

We train the base model for 30k steps ($\sim$983M tokens, batch 64), 4 seeds, and measure branch importance at each checkpoint (mean $\pm$ std, Table 1, Figure 1).

At 98M tokens—the regime of many pilot studies and of our own early experiments—the shared expert carries 81% of the ablation impact and the routed experts look nearly redundant. The crossover to balance happens by $\sim$295M tokens, after which the branches contribute equally through 1B. All four seeds follow the same trajectory with tight error bars; routed specialization grows in parallel (NMI $0.0008 \rightarrow 0.017$). **Conclusions drawn about routed redundancy or shared dominance below a few hundred million tokens do not generalize.**

5 The mechanism: a gradient-exposure race

Why does the shared expert dominate early? It is updated on **every** token; each routed expert is updated on only a $k/E = 2/8 = 25\%$ fraction. The shared expert therefore wins the early optimization race and the residual stream comes to rely on it; the routed experts become load-bearing only once the shared expert saturates.

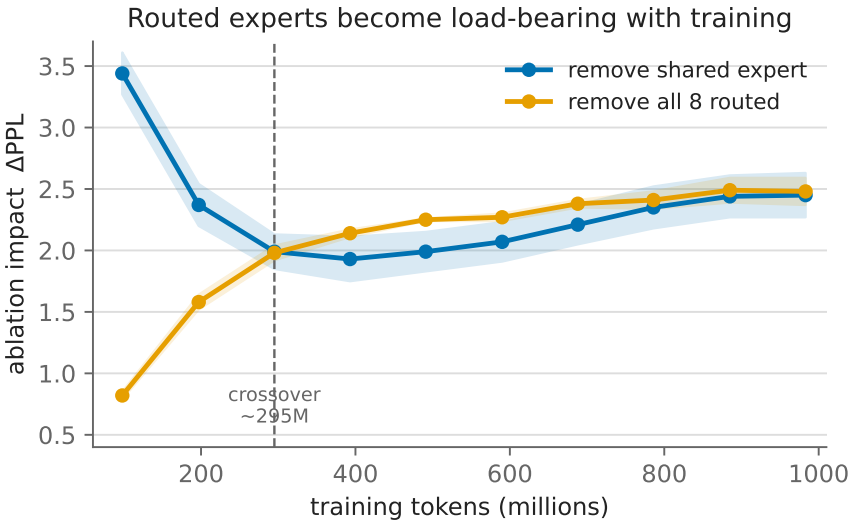


Figure 1: Ablation impact (ΔPPL) of removing the shared expert vs. all 8 routed experts, over a 4-seed constant-batch trajectory (bands = ±1 std). The shared expert dominates early; the branches cross to balance by ∼295M tokens.

Table 2: Shared-expert LR sweep (2 seeds), at 98M tokens.

shared LR	remove routed	remove shared	shared share	PPL
×1.0 (baseline)	+0.85	+3.46	0.80	4.16
×0.3	+1.94	+2.48	0.56	4.10
×0.1	+4.46	+0.72	0.14	4.10

Formalization. Consider one MoE block whose output on token t is

$$y_t = s(x_t) + \sum_{e \in \text{topk}(t)} g_{t,e}\, f_e(x_t), \tag{1}$$

where s is the shared expert and f_e the routed experts. Over a batch of N tokens, the shared parameters receive gradient aggregated over all N tokens, whereas a routed expert e receives gradient only from the $\approx Nk/E$ tokens routed to it. Under a shared learning rate, the shared expert accumulates roughly $E/k\times$ more effective updates per parameter early in training, when the loss gradient is large and the allocation of "who computes what" is still plastic. The residual pathway greedily routes common computation to whichever sub-module reduces loss fastest—the shared expert—further starving the routed experts (a rich-get-richer dynamic). This yields two testable predictions: (i) reducing the shared expert's learning rate should let the routed experts capture computation earlier (confirmed below); and (ii) the crossover token count should grow with E/k (Section 6, only partially confirmed).

We test prediction (i) by placing the shared-expert parameters in their own optimizer group at lr × mult, everything else at the base learning rate (2 seeds; values at 98M tokens, Table 2, Figure 2).

Slowing the shared expert makes the routed branch load-bearing **dramatically earlier**—a 5× increase in routed importance at a *fixed* token budget—shifting the crossover from ∼295M tokens (baseline) to <100M (×0.3) to before training even settles (×0.1, where routed dominate from

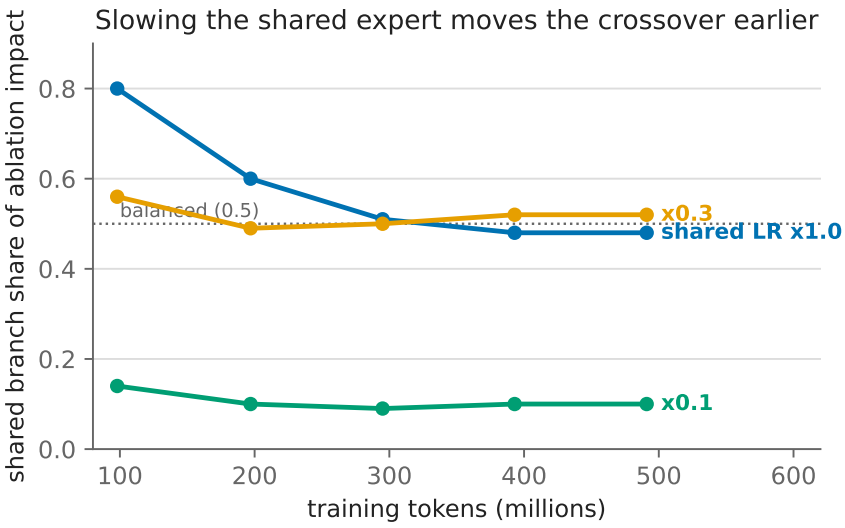


Figure 2: Shared branch's share of ablation impact vs. tokens, for three shared-expert learning-rate multipliers (2 seeds). Slowing the shared expert shifts the routed-activation crossover earlier, dose-dependently—at unchanged perplexity.

the start). The effect is monotonic, dose-dependent, tight error bars. This **refutes autonomous routed maturation**: the routed experts were being out-competed, not merely immature. Crucially, **held-out perplexity is unchanged** across conditions. The shared expert is therefore not *necessary* for this computation; the model defaults to placing it there because that expert trains fastest. A single knob reallocates computation to the routed experts at no quality cost.

Does the mechanism hold at scale? We repeat the shared-LR sweep on a 2×-deeper model (16 layers, 251M parameters, 1 seed; Figure 3). The control reproduces almost exactly: at 98M tokens the shared share drops $0.82 \to 0.64 \to 0.21$ for $\times 1.0/\times 0.3/\times 0.1$ (vs. 0.80/0.56/0.14 at 134M), dose-dependent and at unchanged perplexity—the gradient-race mechanism and its knob are robust to depth. The crossover *timing*, however, is scale-dependent: the deeper baseline only reaches a minimum share of 0.54 (∼491M tokens) and does **not** cross to balance within 1B tokens, the shared expert re-asserting dominance (0.59 at 983M)—the same dip-then-rise seen for larger E (Section 6). Deeper models stay shared-dominated for many more tokens; "enough tokens" for routed balance grows with model size (mechanistically consistent). This strengthens the practical warning but means we observe the crossover *complete* only at our smallest configuration within our 1B-token budget.

6 Scope: the #experts dependence is non-monotonic

The gradient-race picture naively predicts the crossover should scale $\propto E/k$. We test $E \in \{4, 8, 16, 32\}$ (top-2, batch 64, full trajectories) and find this clean law **does not hold** (Table 3, Figure 4).

The dependence is **non-monotonic**, with a "Goldilocks" optimum at E=8: only there do the routed experts fully reach balance. The mechanism explains the high-E side (E=8 → 16 → 32: as each routed expert's gradient share k/E shrinks, the shared expert grows more dominant) but fails at E=4, which stays shared-dominant despite the *highest* per-expert gradient share—implicating

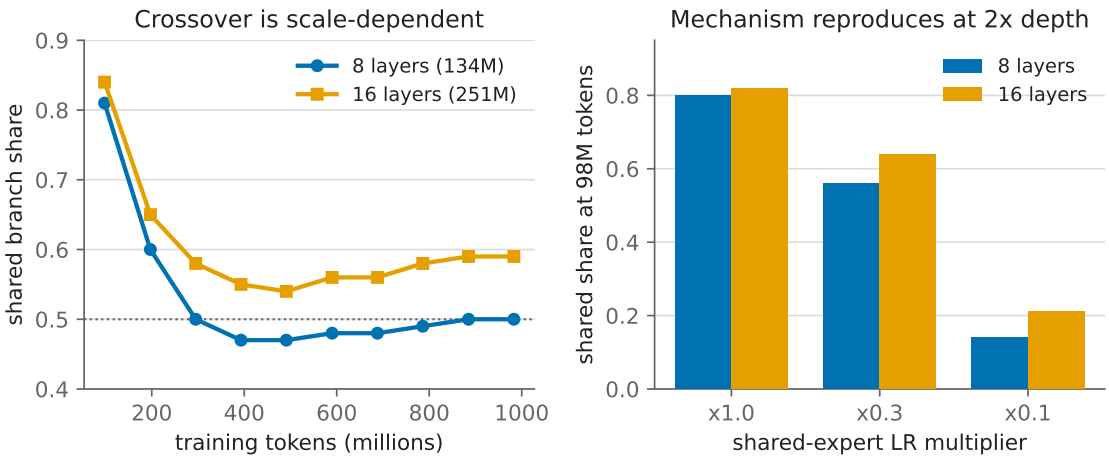


Figure 3: (left) shared branch's share vs. tokens at two depths—the 8-layer model crosses to balance, the 16-layer model approaches but does not within 1B tokens; (right) the shared-LR control reproduces at 2× depth, dropping the share dose-dependently at both scales.

Table 3: #experts sweep (top-2, batch 64): minimum shared share and final PPL.

#experts	min shared share	crosses 0.5?	final PPL
4	0.56	no	3.28
8	0.48	yes	3.00
16	0.53	no	2.94
32	0.55	no	2.90

a second factor, total routed capacity (E=4 has half of E=8's routed parameters). There is a practical tension: more experts improve perplexity (2.90 at E=32) but leave the routed experts individually more under-used.

7 Supporting characterization and retracted findings

What experts specialize by. On models trained without the load-balancing loss (where specialization is strongest), NMI(Expert; axis) is largest for *surface form* (0.050), then domain (0.034), then token frequency (0.028), and essentially zero for sequence position (0.0003). A conditional-MI check shows the axes are separable—domain routing information is fully retained after conditioning on surface class—so domain specialization is genuine, not a surface-form artifact. This reproduces, in a controlled labelled setting, prior reports that experts organize more by token/surface than by topic.

Load balancing controls specialization. Sweeping the auxiliary loss coefficient $\{0, 10^{-2}, 10^{-1}\}$ moves domain NMI monotonically (0.034/0.0085/0.0010; all pairwise permutation tests significant) and load imbalance in lockstep, at no perplexity cost (Figure 5).

Shared expert enables specialization. With balancing off, routed domain-NMI is higher *with* the shared expert (0.034) than without (0.015, $p = 0.029$): the shared expert acts as a common-

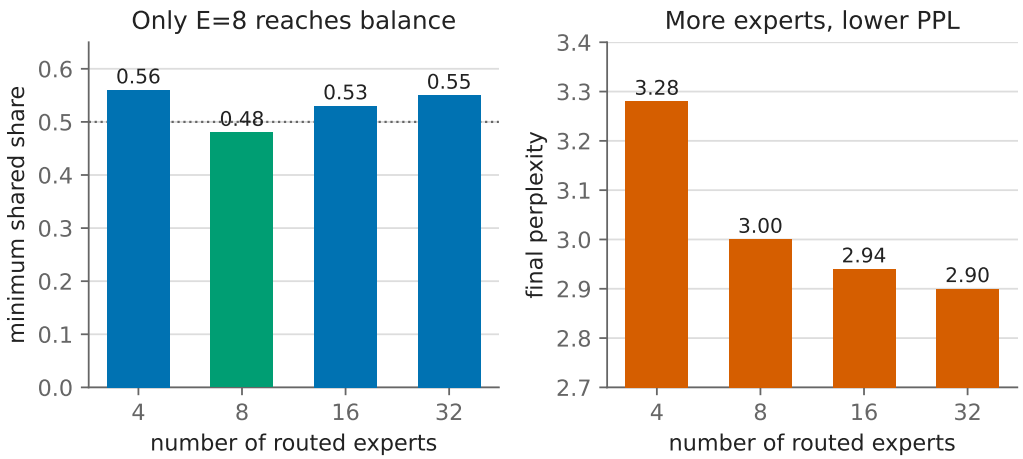


Figure 4: (left) minimum shared share vs. number of routed experts—only E=8 crosses to balance; (right) more experts monotonically improve perplexity. Engagement and perplexity are optimized at different granularities.

capability backstop, consistent with the DeepSeekMoE intuition.

Two retracted findings. (a) At 3 seeds, removing the shared expert appeared to sharpen routed specialization; at 4 seeds this collapsed ($p = 0.54$) and, once balancing is off, the sign reverses. (b) We hypothesized the shared monopoly was a *pathology*; a per-token shared-branch dropout breaks the monopoly entirely (shared share $\rightarrow 0.01$) but yields no perplexity gain and no specialization gain—consistent with Section 5's conclusion that the shared/routed split is a near-neutral reparametrization. We report both as negatives.

Distillation (preliminary). On-the-fly distillation from a Mistral-7B teacher, compared to a batch/step-matched from-scratch control at 98M tokens, appears to engage the routed branch more (remove-routed $+1.03 \rightarrow +1.74$) but at a small next-token PPL cost and from a single seed; not established.

8 Practical recommendations

1. **Do not diagnose expert redundancy on an undertrained model.** Below a few hundred million tokens the shared expert dominates for optimization reasons, not architectural ones; measure branch importance as a trajectory, not at a single early checkpoint.

2. **To engage the routed experts sooner, slow the shared expert** (e.g. a $\times 0.3$ LR multiplier). This reallocates computation to the routed branch at no measured perplexity cost—valuable when routed experts are the object of study or when balanced load is desired.

3. **Pick the number of experts for engagement, not just perplexity.** More experts improved perplexity but left the routed experts individually more under-used; engagement was maximized at an intermediate count (E=8).

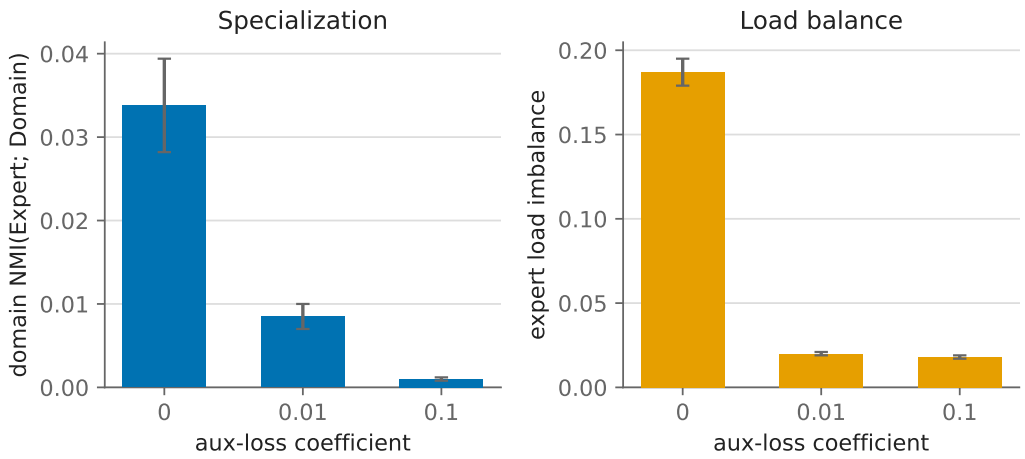


Figure 5: Domain specialization (left) and expert load imbalance (right) vs. the auxiliary load-balancing coefficient (4 seeds, ±1 std). Both fall monotonically as balancing strengthens.

4. **Report seeds and use matched batch sizes.** Both the sign of a specialization effect and the magnitude of branch-importance metrics were sensitive to seed and batch size at fixed tokens.

9 Limitations

Pilot scale (134M parameters, ≤1B tokens, 8 experts, 512 context, a single hardware configuration and data mix). Held-out evaluation uses a skip-offset of the training streams rather than a fully disjoint split. Held-out NMI and branch-importance magnitudes are sensitive to batch size at fixed token count, so cross-experiment magnitude comparisons should use matched batches; the within-experiment, matched-batch comparisons and the token trajectories are the reliable ones. Distillation results are single-seed. Whether the ∼300M-token crossover and the gradient-race control persist at larger scale is untested.

10 Conclusion

In small shared-expert MoEs, the routed experts are not redundant—they are out-competed by the shared expert in a gradient-exposure race that resolves with training and that a single learning-rate knob controls at no quality cost. The practical takeaways are concrete: do not diagnose expert redundancy on an undertrained MoE, and if you want the routed experts engaged sooner, slow the shared expert. The methodology—a cheap, controllable, domain-labelled testbed with causal branch ablation, seed replication, and the willingness to retract—is offered alongside the findings. All code, configs, and the analysis toolkit run on a single GPU and are released with the paper.

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